# Project Initialization and Planning Phase

| | |
|---|---|
| Date | 20 October 2026 |
| Team ID | En25cs3010862 |
| Project Title | Health360: Comprehensive Healthcare Risk & Disease Analytics |
| Maximum Marks | 3 Marks |

## Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

| **Project Overview** | |
|---|---|
| Objective | To build Health360, an AI-driven and IoT-integrated healthcare analytics platform that continuously monitors patient vitals, performs multi-disease risk prediction, and alerts medical professionals to enable early, proactive interventions. |
| Scope | Includes real-time data ingestion from wearable sensors, machine learning models for early disease detection (cardiovascular, diabetes, respiratory), interactive web/mobile dashboards for patients and doctors, and HIPAA/GDPR-compliant encrypted cloud storage. Excludes direct surgical/robotic interventions. |
| **Problem Statement** | |
| Description | Traditional healthcare systems are predominantly reactive, fragmented, and burdened by data silos. Patients with chronic conditions lack continuous monitoring between clinic visits, leading to delayed diagnoses, preventable complications, and high hospital readmission rates. |
| Impact | Solving this problem transitions care from reactive to preventive, significantly reducing critical medical emergencies, lowering healthcare costs for patients, minimizing clinical workload, and improving overall survival and quality of life. |
| **Proposed Solution** | |

| | |
|---|---|
| Approach | Health360 employs a 5-layer IoT-Cloud-AI architecture: 1) Non-invasive wearable sensing, 2) Encrypted MQTT/REST data transmission, 3) Automated data cleaning and ETL pipelines, 4) Predictive analytics via supervised ML models (Scikit-Learn/TensorFlow), and 5) Real-time alert notifications and interactive React web/mobile dashboards. |
| Key Features | Continuous vital sign tracking (Heart Rate, SpO2, Temperature); Automated ML-based disease risk scoring; Real-time emergency alert triggers for doctors and caregivers; Role-based dashboards for patients, doctors, and admins; Secure, encrypted EHR data management. |

**Resource Requirements**

| **Resource Type** | **Description** | **Specification/Allocation** |
|---|---|---|
| **Hardware** | | |
| Computing Resources | CPU / GPU specs for model training & API service | 8-Core Intel/AMD CPU, 1 x NVIDIA RTX 3080 / T4 GPU |
| Memory | System RAM for pipeline processing & data serving | 16 GB DDR4 RAM |
| Storage | Cloud / Local SSD for EHR storage, ML models, and logs | 512 GB NVMe SSD + AWS S3 Storage |
| **Software** | | |
| Frameworks | Backend REST API & Web Frameworks | Flask / FastAPI, React.js |
| Libraries | Data analytics, machine learning, and visualization | scikit-learn, pandas, numpy, TensorFlow, Chart.js |
| Development Environment | IDE, testing environment, and version control | VS Code, Jupyter Notebook, Git, GitHub, Postman |
| **Data** | | |
| Data | Synthetic & public medical datasets (EHR, vitals, risk indicators) | MIMIC-III / Kaggle Healthcare Datasets (~50,000 anonymized patient records, CSV format) |